6GGW / NetSwitch — numbered math index

Every math function in the suite, numbered `M1…Mn`, with its file, signature, and what it computes. This index is the authoritative numbering; the newest source files also carry matching `// [Mxx]` tags at the function (see `ddtm`, `approve`, `netswitch-sql/scaling`). Ask and I'll stamp the tags into any other file too.

Compute kernel — `dramm/`, `loopbench/`, `approve/`

#	File · function	Computes
M1	`dramm/cpu_benchmark.cpp · step(x,i)`	one DRAMM step: `v=ln(x+1)+1; acc=Σ_{k=1..40} sqrt(v/k); return 1/(acc/40 + ((i&7)+1))`
M2	`loopbench/ggw_loopbench.cpp · step_opt / dramm_sum_const`	closed form: `Σ sqrt(v/k)=sqrt(v)·C`, `C=Σ1/√k=11.267648377839` (precompute once)
M3	`approve/ggw_approve.cpp · workload(iters)`	deterministic iterate of M2's `step_opt` → the reproducible result `0.120493199688742`
M4	`approve/ggw_approve.cpp · result_hash(r)`	64-bit fmix of the IEEE-754 bits (murmur finalizer) — same binary ⇒ same hash
M5	`approve/ggw_approve.cpp · approval_ratio(a,b)`	`SQRT((a/b + b/a)/2)` — 1.0 iff timings equal, grows on divergence

Transform / scaling — `ddtm/`, `netswitch-sql/scaling/`

#	File · function	Computes
M6	`ddtm/ggw_ddtm.cpp · dct2`	forward 8×8 DCT-II with `C(0)=√(1/N)`, else `√(2/N)`
M7	`ddtm/ggw_ddtm.cpp · idct2`	inverse 8×8 DCT-III (reconstruct the block)
M8	`ddtm/ggw_ddtm.cpp · (powered split)`	`|C[u,v]| ≥ 0.01·peak` → powered; else unpowered
M9	`ddtm/ggw_ddtm.cpp · entropy(q)`	Shannon entropy of quantized magnitudes, in **bits** and **nats (ln)**
M10	`ddtm/ggw_ddtm.cpp · (requant + PSNR)`	`q=round(C/Q)`, `C'=q·Q`, `PSNR=10·log10(range²/MSE)`
M11	`ddtm/ggw_ddtm.cpp · (best scaler)`	argmax of `PSNR / (bits/64)` above a 30 dB floor
M12	`ddtm/ggw_ddtm.cpp · (MHz banding)`	spectral efficiency `bits·blocks_per_s / bandwidth` → Mbps per MHz
M13	`scaling/ggw_scaling.sql · dbo.ddtm_constant()`	precompute-once constant `-0.090908889` (LLOG)
M14	`scaling/ggw_scaling.sql · dbo.ddtm_constant_calc(...)`	`LOG10( SQRT((T1·MG(7/9)·LogLogRad)/POWER(POWER(T2,T1),MG(1/6))) / 120W )`
M15	`scaling/ggw_scaling.sql · dbo.hyper_downscale(c,f)`	`c/f` — the hypervisor downscaler
M16	`scaling/ggw_scaling.sql · dbo.thrice_sinwave(t,k,…)`	`|k|·Σ_{i=1..3} aᵢ·sin(2π·fᵢ·t)`
M17	`scaling/ggw_scaling.sql · dbo.lnlog(x)`	`LOG(LOG(x))` — ln of ln; `lnlog(e^e)=1`
M18	`scaling/ggw_scaling.sql · dbo.binary_lnlog(w)`	hex bits of `w` plus its lnlog
M19	`scaling/ggw_scaling.sql · dbo.choose_best_scaler`	DB-side max `PSNR/(bits/64)` above floor

Media — `voice-edge/`, `stream-qc/`, `stream-ctl/`, `ipr/`

#	File · function	Computes
M20	`voice-edge/ggw_voice_edge.cpp · ulaw(pcm)`	G.711 μ-law encode (sign, 8 exponent bands, mantissa)
M21	`voice-edge/ggw_voice_edge.cpp · make_rtp(...)`	440 Hz sine → PCMU, RFC 3550 RTP header pack
M22	`stream-qc/ggw_streamqc.cpp · PSNR`	`10·log10(255²/MSE)` fidelity in dB
M23	`stream-qc/ggw_streamqc.cpp · blockiness`	Wang/Bovik 8×8-grid pixelation index (dB), no reference
M24	`stream-ctl/ggw_streamctl.cpp · battery (RKF45)`	RC saturation ODE by Runge–Kutta–Fehlberg 4(5)
M25	`stream-ctl/ggw_streamctl.cpp · battery (Laplace)`	analytic first-order check; agrees with M24 to ~1e-15
M26	`stream-ctl/ggw_streamctl.cpp · pipe capacity`	`bandwidth × MHz` spectral efficiency, −3% floor tier pick
M27	`stream-ctl/ggw_streamctl.cpp · split`	cut points at 1 2 7 15 40 59 80 %, byte-exact rejoin
M28	`ipr/ggw_ipr.cpp · depth ratioing`	far 0.5× / close 1.0× inset scale, inside-only clamp

Measure — `sysmon/`, `cputimer/`, `diskbench/`, `thermal/`

#	File · function	Computes
M29	`sysmon/ggw_sysmon.cpp · measure_cpu_ops_bln`	per-core FMA ops ÷ wall time → billions ops/s
M30	`sysmon/ggw_sysmon.cpp · measure_fps`	offscreen frames rendered ÷ time → FPS
M31	`cputimer/cpu_mhz_timer.h · calibrated RDTSC`	CPU timebase MHz from calibrated cycle count
M32	`cputimer/ggw_radiosig.cpp · reduction chain`	antenna/box/delay/cpu √47.9985 reduction at 1 Hz
M33	`diskbench/ggw_diskbench.cpp · IOPS/MB-s/latency`	fsync-honest seq/random throughput + percentiles
M34	`thermal/ggw_thermal.cpp · trip headroom`	per-part temp vs trip, throttle-swing learn

Control — `server-cpp/`, `registrar/`, `backup-codes/`, `secure/`

#	File · function	Computes
M35	`server-cpp/ggw_server.cpp · rerouting cost`	`cost = RTT × (1 + load/100)`, least-cost reachable POP
M36	`server-cpp/ggw_server.cpp · distance`	`d = 0.0001607 m/ps × TCP-handshake RTT` → signal-path km
M37	`backup-codes/ggw_backupcodes.cpp · sha256`	FIPS 180-4 SHA-256 (also in registrar, voice-edge)
M38	`backup-codes/ggw_backupcodes.cpp · code gen`	Crockford base32, `byte%32` unbiased, 50 bits/code
M39	`registrar/ggw_registrar.cpp · prefix_to_mask`	CIDR prefix → dotted netmask
M40	`secure/ggw_secure.cpp · EC-256 / RSA handshake`	TLS 1.3 key exchange + cert verify (OpenSSL)

Total: **40 math functions** across 20 source files. The three newest modules (`ddtm`, `approve`, `scaling`) carry inline `// [Mxx]` tags matching this table; say the word and I’ll stamp the rest.